

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
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Comprehensive Examination

EC-3 Regular Solution

Course No. : AIMLCZC416
Course Title : Mathematical Foundations for Machine Learning
Nature of Exam : Open Book
Weightage : 40%

Q1. Consider the matrix $A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$.

(a) Find all the eigen values and eigen vectors of A. [5M]

To find the eigen values of A we have to solve the equation $|A - \lambda I| = 0$.

$$A - \lambda I = \begin{bmatrix} 3 - \lambda & 1 & 0 \\ 1 & 3 - \lambda & 0 \\ 0 & 0 & 4 - \lambda \end{bmatrix}$$

$$\begin{aligned} \text{Now } |A - \lambda I| &= (4 - \lambda) \det \begin{bmatrix} 3 - \lambda & 1 \\ 1 & 3 - \lambda \end{bmatrix} = (4 - \lambda)[(3 - \lambda)^2 - 1] \\ &= (4 - \lambda)(9 - 6\lambda + \lambda^2 - 1) \\ &= (4 - \lambda)(\lambda^2 - 6\lambda + 8) \\ &= (4 - \lambda)(\lambda - 2)(\lambda - 4) \quad [1M] \end{aligned}$$

Hence eigenvalues are: $\boxed{\lambda_1 = 2, \lambda_2 = 4, \lambda_3 = 4}$ [1M]

Step 2: Find Eigenvectors

For $\lambda = 2$

$$A - 2I = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Equations:

$$\begin{aligned} x + y &= 0 \\ z &= 0 \end{aligned}$$

So

$$x = -y \quad [0.5M]$$

Eigenvector:

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad [1M]$$

For $\lambda = 4$

$$A - 4I = \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Equation:

$$-x + y = 0 \Rightarrow x = y \quad [0.5M]$$

z is free.

Two independent eigenvectors:

$$v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad [1M]$$

(b) Diagonalization [2M]

Form matrix P using eigenvectors as columns:

$$P = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad [1M]$$

Diagonal matrix:

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \quad [1M]$$

Thus,

$$\boxed{AP = PD}$$

(c) Singular Values [1M]

Matrix given:

$$B = \begin{bmatrix} \sqrt{3} & \frac{1}{\sqrt{3}} & 0 \\ 0 & \sqrt{\frac{8}{3}} & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Since B is upper triangular, compute

$$B^T B = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

This is exactly matrix A .

Singular values = $\sqrt{\text{eigenvalues of } B^T B}$.

Eigenvalues of A are:

$$2, 4, 4$$

Hence singular values:

$$\boxed{\sqrt{2}, 2, 2} \quad [1M]$$

(b) Using (a) find a diagonal matrix D and an invertible matrix P such that $AP = PD$. [2M]

(c) Find the singular values of the matrix $\begin{bmatrix} \sqrt{3} & \frac{1}{\sqrt{3}} & 0 \\ 0 & \sqrt{\frac{8}{3}} & 0 \\ 0 & 0 & 2 \end{bmatrix}$. [1M]

Q2. Let $v_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ -1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 2 \\ 3 \\ 4 \end{bmatrix}$, $v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, and $v_4 = \begin{bmatrix} -6 \\ 3 \\ -3 \\ -3 \end{bmatrix}$,

(a) Determine whether the vectors v_1, v_2, v_3, v_4 are linearly dependent or linearly independent in $\mathbb{R}^4$. Justify your answer. [3M]

(b) Prove that $U = \text{span} \{v_1, v_2, v_3, v_4\}$ is a subspace of $\mathbb{R}^4$. Also find a basis and dimension of U . [3M]

(c) What is the rank of the matrix $B = \begin{bmatrix} -3 & 0 & 0 & -6 \\ 1 & 2 & 1 & 3 \\ 0 & 3 & 0 & -3 \\ -1 & 4 & 0 & -3 \end{bmatrix}$. Justify your answer. Also find the determinant of the matrix B . [2M]

We form the matrix B using the given vectors as columns:

$$B = \begin{bmatrix} -3 & 0 & 0 & -6 \\ 1 & 2 & 1 & 3 \\ 0 & 3 & 0 & -3 \\ -1 & 4 & 0 & -3 \end{bmatrix} \quad [1M]$$

Step 1: Row Reduction to Echelon Form

Interchange $R_1 \leftrightarrow R_2$ (to get leading 1)

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ -3 & 0 & 0 & -6 \\ 0 & 3 & 0 & -3 \\ -1 & 4 & 0 & -3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 3R_1$$

$$R_4 \rightarrow R_4 + R_1$$

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 6 & 3 & 3 \\ 0 & 3 & 0 & -3 \\ 0 & 6 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 6 & 3 & 3 \\ 0 & 0 & -\frac{3}{2} & -\frac{9}{2} \\ 0 & 0 & -2 & -3 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - \frac{4}{3}R_3$$

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 6 & 3 & 3 \\ 0 & 0 & -\frac{3}{2} & -\frac{9}{2} \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad [1M]$$

Since all the columns in the echelon form have pivot element, the vectors v_1, v_2, v_3, v_4 are linearly independent. [1M]

To prove U is closed under addition:

Take any two vectors in U :

$$\begin{aligned} u &= a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4 \\ w &= b_1v_1 + b_2v_2 + b_3v_3 + b_4v_4 \end{aligned}$$

where $a_i, b_i \in \mathbb{R}$.

Now add them:

$$u + w = (a_1v_1 + \cdots + a_4v_4) + (b_1v_1 + \cdots + b_4v_4)$$

Group like terms:

$$u + w = (a_1 + b_1)v_1 + (a_2 + b_2)v_2 + (a_3 + b_3)v_3 + (a_4 + b_4)v_4$$

Since $a_i + b_i$ are real numbers, this is again a linear combination of v_1, v_2, v_3, v_4 .

Therefore,

$$\boxed{u + w \in U} \quad [1M]$$

To prove U is closed under scalar multiplication:

Let $u \in U$:

$$u = a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4$$

Let $c \in \mathbb{R}$.

Multiply:

$$cu = c(a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4)$$

Distribute scalar:

$$cu = (ca_1)v_1 + (ca_2)v_2 + (ca_3)v_3 + (ca_4)v_4$$

Since ca_i are real numbers, this is again a linear combination of v_1, v_2, v_3, v_4 .

Hence,

$$\boxed{cu \in U} \quad [1m]$$

Note: If they simply write U is closed under vector addition and scalar multiplication U is a subspace give 1 M instead of 2.

Since v_1, v_2, v_3, v_4 are linearly independent and span U they form a basis of U {0.5M]

Dimension of U = 4.

(c) Since the number of non-zero rows in the echelon form of B are 4, rank of B = 4. [1M]

Determinant of B = $(-1) * 1 * 6 * (-3/2) * 3 = 27$ [1M]

Q3. Let $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.

An inner product on $\mathbb{R}^2$ is defined by $\langle u, v \rangle = u^T A v$, for $u, v \in \mathbb{R}^2$.

The norm induced by this inner product is $\|x\| = \sqrt{\langle x, x \rangle}$.

Assume that a signal tower is located at the origin $(0, 0)$. The cost of placing a receiver at a point $(x, y) \in \mathbb{R}^2$ is given by $C(x, y) = 2 \| (x, y) \|^2$. Suppose the receiver must be placed in the region

$$R = \{(x, y) \in \mathbb{R}^2 : x + y - 1 \geq 0, x + y - 3 \leq 0\}.$$

(a) State and prove the necessary properties of the matrix A such that $\langle u, v \rangle = u^T A v$ defines an inner product on $\mathbb{R}^2$. [2M]

The properties of A should be A is symmetric and A is positive definite. [0.5M]

Clearly $A = A^T$. [0.5M]

In order to prove A is positive definite we shall prove that determinant of every principal minor of $A > 0$.

Clearly $\det [3] = 3 > 0$ and $\det \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} = 9 - 1 = 8 > 0$. [1M]

Note: If they prove $X^T A X > 0$ for any $X \neq 0$ in $\mathbb{R}^2$ please award full mark.

(b) Formulate the constrained optimization problem to determine the point in the region R at which the cost function $C(x, y)$ is minimized. [1M]

Norm:

$$\|X\|^2 = X^T A X$$

For $X = \begin{bmatrix} x \\ y \end{bmatrix}$:

$$\begin{aligned} X^T A X &= \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= 3x^2 + 2xy + 3y^2 \end{aligned}$$

Primal Problem:

$$\boxed{\min_{x,y} 6x^2 + 4xy + 6y^2}$$

subject to

$$x + y \geq 1, x + y \leq 3 \quad [1M]$$

(c) Let $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ be the Lagrange multipliers corresponding to the constraint $x + y - 1 \geq 0$ and $x + y - 3 \leq 0$ respectively. Formulate the Lagrangian function and hence write the corresponding Lagrangian dual problem. [2M]

Rewrite constraints as:

$$\begin{aligned} g_1(x, y) &= 1 - x - y \leq 0 \\ g_2(x, y) &= x + y - 3 \leq 0 \end{aligned}$$

with $\lambda_1, \lambda_2 \geq 0$.

Lagrangian

$$L(x, y, \lambda_1, \lambda_2) = 6x^2 + 4xy + 6y^2 + \lambda_1(1 - x - y) + \lambda_2(x + y - 3) \quad [1M]$$

Dual Function

$$D(\lambda_1, \lambda_2) = \min_{x, y} L(x, y, \lambda_1, \lambda_2)$$

Dual Problem

$$\boxed{\max D(\lambda_1, \lambda_2)} \quad \text{subject to } \lambda_1 \geq 0 \text{ and } \lambda_2 \geq 0 \quad [1M]$$

(d) Verify that the Karush–Kuhn–Tucker (KKT) conditions are satisfied at $(x, y, \lambda_1, \lambda_2) = \left(\frac{1}{2}, \frac{1}{2}, 8, 0\right)$. [3M]

1.

$$g_1\left(\frac{1}{2}, \frac{1}{2}\right) = 1 - \frac{1}{2} - \frac{1}{2} = 0 \leq 0$$

$$g_2\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2} + \frac{1}{2} - 3 = -2 \leq 0 \quad [0.5M]$$

$$2. \quad \lambda_1 = 8 \geq 0 \text{ and } \lambda_2 = 0 \geq 0 \quad [0.5M]$$

$$3. \quad 8 * g_1\left(\frac{1}{2}, \frac{1}{2}\right) = 8 * 0 = 0$$

$$0 * g_2\left(\frac{1}{2}, \frac{1}{2}\right) = 0 * (-2) = 0 \quad [0.5M]$$

4. Compute gradients.

$$\nabla C = \begin{bmatrix} 12x + 4y \\ 4x + 12y \end{bmatrix}$$

At $(1/2, 1/2)$:

$$\nabla C = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

Gradient of constraints:

$$\nabla g_1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$\nabla g_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad [0.5M]$$

Stationarity condition:

$$\begin{aligned} \nabla C + \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2 &= 0 \\ &= \begin{bmatrix} 8 \\ 8 \end{bmatrix} + 8 \begin{bmatrix} -1 \\ -1 \end{bmatrix} + 0 \\ &= \begin{bmatrix} 8 \\ 8 \end{bmatrix} + \begin{bmatrix} -8 \\ -8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad [1M] \end{aligned}$$

Hence KKT conditions are satisfied at $(\frac{1}{2}, \frac{1}{2}, 8, 0)$.

Q4. A retail shop wants to predict their sales corresponding to the money they spent on advertisement using the following data. A linear regression model with bias is assumed:

$$\hat{y}_i = w_0 + w_1 x_i$$

Money spent for advertisement in ten thousands (x)	Sales in lacs (y)
5	3
6	5

The model parameters are learned by minimizing the Squared Error loss function $J(w_0, w_1) = \frac{1}{2} \sum_{i=1}^2 (\hat{y}_i - y_i)^2$ using gradient descent.

(a) Formulate the Squared Error loss function $J(w_0, w_1)$ explicitly for the given dataset. Compute the Hessian matrix of J with respect to w_0 and w_1 , and use it to prove that J is a convex function of the parameters (w_0, w_1) . [3M]

$$J(w_0, w_1) = \frac{1}{2} [(w_0 + 5w_1 - 3)^2 + (w_0 + 6w_1 - 5)^2] \quad [0.5M]$$

Step 2: Compute Hessian

First compute partial derivatives.

$$\begin{aligned} \frac{\partial J}{\partial w_0} &= (w_0 + 5w_1 - 3) + (w_0 + 6w_1 - 5) \\ &= 2w_0 + 11w_1 - 8 \end{aligned}$$

$$\begin{aligned} \frac{\partial J}{\partial w_1} &= 5(w_0 + 5w_1 - 3) + 6(w_0 + 6w_1 - 5) \\ &= 11w_0 + 61w_1 - 45 \end{aligned}$$

Second derivatives (Hessian)

$$H = \begin{bmatrix} \frac{\partial^2 J}{\partial w_0^2} & \frac{\partial^2 J}{\partial w_0 \partial w_1} \\ \frac{\partial^2 J}{\partial w_1 \partial w_0} & \frac{\partial^2 J}{\partial w_1^2} \end{bmatrix}$$

$$H = \begin{bmatrix} 2 & 11 \\ 11 & 61 \end{bmatrix} \quad [0.5M]$$

Step 3: Check Positive Definiteness

Principal minors:

$$2 > 0$$

$$\det(H) = 2(61) - 11^2 = 122 - 121 = 1 > 0 \quad [2M]$$

Since leading principal minors > 0 ,

H is positive definite

Therefore,

$J(w_0, w_1)$ is convex

(b) Compute the gradient of the Squared Error loss function: $\nabla J = \begin{bmatrix} \frac{\partial J}{\partial w_0} \\ \frac{\partial J}{\partial w_1} \end{bmatrix}$. [1M]

$$J = \begin{bmatrix} 2w_0 + 11w_1 - 8 \\ 11w_0 + 61w_1 - 45 \end{bmatrix} \quad [1M]$$

(c) Taking the initial parameter values $w_0^{(0)} = 0$, $w_1^{(0)} = 0$, perform two iterations of gradient descent using the inverse decaying learning rate schedule with initial learning rate $\eta_0 = 0.1$, $k=0.4$, where $t=1, 2, \dots$ denote the iteration index. Show the updated parameter values after each iteration. [4M]

Learning rate schedule (inverse decay):

$$\eta_t = \frac{\eta_0}{1 + kt} = \frac{0.1}{1 + 0.4t}$$

Iteration 1 (t=1)

$$\eta_1 = \frac{0.1}{1.4} = 0.07143 \quad [0.5M]$$

Initial:

$$w_0^{(0)} = 0, w_1^{(0)} = 0$$

Gradient at (0,0):

$$\nabla J = \begin{bmatrix} -8 \\ -45 \end{bmatrix} \quad [0.5M]$$

Update rule:

$$w^{(1)} = w^{(0)} - \eta_1 \nabla J$$

$$w_0^{(1)} = 0 - 0.07143(-8) = 0.5714 \quad [0.5M]$$

$$w_1^{(1)} = 0 - 0.07143(-45) = 3.2144 \quad [0.5M]$$

Iteration 2 (t=2)

$$\eta_2 = \frac{0.1}{1.8} = 0.05556 \quad [0.5M]$$

Compute gradient at (0.5714, 3.2144):

$$\begin{aligned}
\frac{\partial J}{\partial w_0} &= 2(0.5714) + 11(3.2144) - 8 \\
&= 1.1428 + 35.3584 - 8 = 28.5012 \\
\frac{\partial J}{\partial w_1} &= 11(0.5714) + 61(3.2144) - 45 \\
&= 6.2854 + 196.0784 - 45 = 157.3638 \text{ [0.5M]}
\end{aligned}$$

$$\text{Gradient} \approx \begin{bmatrix} 28.5012 \\ 157.3638 \end{bmatrix}$$

Update:

$$\begin{aligned}
w_0^{(2)} &= 0.5714 - (0.05556 * 28.5012) \\
&= 0.5714 - 1.5835 = -1.0121 \text{ [0.5M]}
\end{aligned}$$

$$\begin{aligned}
w_1^{(2)} &= 3.2144 - (0.05556 * 157.3638) \\
&= 3.2144 - 8.7431 = -5.5287 \text{ [0.5M]}
\end{aligned}$$

Q5. Consider the following training data set in R^2 .

Point	x_1	x_2	class
X_1	2	2	+1
X_2	4	4	+1
X_3	2	0	-1
X_4	4	0	-1

(a) Let $w = (w_1, w_2)^T \in \mathbb{R}^2$ denote the weight vector and $b \in \mathbb{R}$ the bias term.

Formulate the hard-margin Support Vector Machine (SVM) primal optimization problem for the given dataset. [2M]

For hard-margin SVM, the primal optimization problem is:

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to constraints:

$$y_i(w^T x_i + b) \geq 1, i = 1, \dots, 4$$

Thus the Hard Margin problem becomes

$$\min_{w,b} \frac{1}{2}(w_1^2 + w_2^2) \quad [1M]$$

Subject to

$$2w_1 + 2w_2 + b \geq 1$$

$$4w_1 + 4w_2 + b \geq 1$$

$$-(2w_1 + b) \geq 1$$

$$-(4w_1 + b) \geq 1 \quad [1M]$$

(b) show that the line $x_2 = 1$ is a separating hyper plane for the given data. [2M]

$$\text{Equation: } x_2 - 1 = 0$$

Standard form:

$$w = (0,1), b = -1$$

Decision function:

$$f(x) = x_2 - 1$$

Check all points:

Positive class:

$$X_1: 2 - 1 = 1 > 0$$

$$X_2: 4 - 1 = 3 > 0 \quad [1M]$$

Negative class:

$$X_3: 0 - 1 = -1 < 0$$

$$X_4: 0 - 1 = -1 < 0 \quad [1M]$$

All positive points lie on one side and negative on the other.

$$x_2 = 1 \text{ is a separating hyperplane}$$

(c) show that the line $x_2 - x_1 = -1$ is a separating hyper plane for the given data. [2M]

Rewriting the given equation $x_2 - x_1 + 1 = 0$.

$$\text{So } w = (-1,1), b = 1$$

Decision function:

$$f(x) = x_2 - x_1 + 1$$

Check points:

$$X_1: 2 - 2 + 1 = 1 > 0$$

$$X_2: 4 - 4 + 1 = 1 > 0 \text{ [1M]}$$

$$X_3: 0 - 2 + 1 = -1 < 0$$

$$X_4: 0 - 4 + 1 = -3 < 0 \text{ [1M]}$$

Correct separation.

$x_2 - x_1 = -1$ is a separating hyperplane

(d) Compute the margin of each hyperplane obtained in parts (b) and (c). Determine which hyperplane provides the larger margin and justify your answer. [2M]

Margin formula:

$$\text{Margin} = \frac{1}{\|w\|}$$

Hyperplane 1: $x_2 = 1$

$$w = (0, 1)$$

$$\|w\| = 1$$

$$\text{Margin} = \frac{1}{1} = 1$$

Hyperplane 2: $x_2 - x_1 = -1$

$$w = (-1, 1)$$

$$\|w\| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$\text{Margin} = \frac{1}{\sqrt{2}} = 0.7071$$

Comparison $1 > 0.7071$.

Therefore,

$x_2 = 1$ provides the larger margin

Hence it is the better SVM separator.
